

Amazon Laptop Market Analysis Report

Objective

The goal of this project is to collect laptop data from Amazon India using web scraping techniques, clean and structure the dataset, and perform exploratory data analysis to identify market trends such as pricing, specifications, and brand performance.

Data Collection

Laptop product data was scraped from Amazon India search results using Python. The scraper collected more than 100 laptop listings across multiple pages.

Extracted Attributes

- Product Title
- Price
- Rating
- Number of Reviews
- Brand
- RAM
- Storage Type
- Processor
- Windows Version
- Screen Size
- Color

The raw dataset was stored as:

amazon_laptop_raw.csv

After cleaning and preprocessing:

amazon_laptop_cleaned.csv

Data Cleaning & Preparation

The following preprocessing steps were performed:

1. Removed duplicate products
 2. Converted price and rating to numeric values
 3. Extracted specifications from product titles using regular expressions
 4. Removed missing and invalid entries
 5. Standardized RAM and storage formats
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Data Analysis & Findings

1. Brand with Highest Average Price

Brands with premium configurations such as high RAM and SSD capacity showed the highest average pricing.

This indicates premium branding and performance-oriented targeting.

2. Brand with Highest Average Rating

Certain brands consistently maintained higher customer satisfaction ratings due to reliability and build quality.

3. Top 5 Most Reviewed Laptops

Products with high review counts typically belonged to mid-range budget categories, suggesting these are the most popular purchase segment.

4. Price Distribution

The majority of laptops were concentrated in the ₹40,000 – ₹80,000 price range, indicating this is the most competitive and demanded segment in the Indian market.

5. Price vs Rating Relationship

Correlation between price and rating was weak.
Higher price does not guarantee higher rating — customers value reliability over cost.

6. RAM Distribution

Most laptops were:

- 8GB RAM (majority)
- 16GB RAM (second most common)

This shows 8GB is still the market standard configuration.

7. Most Common Screen Size

The most frequent screen size was around **15.6 inches**, which is the standard laptop form factor.

8. Laptops Under ₹50,000

A significant portion of listings belonged to budget category under ₹50,000, indicating strong demand for affordable laptops among students and office users.

Conclusion

The Amazon India laptop market is dominated by mid-range laptops with 8GB RAM and 15.6-inch screens.

Consumers prioritize reliability and value for money rather than premium specifications.

Budget and mid-range segments form the highest demand, while premium laptops exist but serve niche users.

Final Output Files

- amazon_laptop_raw.csv

- amazon_laptop_cleaned.csv
- analysis_report.pdf